

torch-spyre Ecosystem Health Report

Date: 2026-05-22 | Pod: rganti-spyre-main | PyTorch: 2.11.0+cpu

3/3 BUILD PASS	FAIL NIGHTLY CI	50 PRS MERGED (7D)	30 OPEN BUGS
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Build Status

Component	Commit	Status	Notes
deeptools	b7f6fdbfa7 — Update dxp.cpp	PASS	cmake + make - j16 install
flex	8f06481 — remove default directive from FlexAllocator::allocate (#1070)	PASS	cmake + make - j16 install
torch-spyre	7679a4f — Implement working set reduction hints (#2226)	PASS	uv sync editable install (~1m 19s)

CI Status

Workflow	Latest	Status	Trend
Spyre Hardware Tests	2026-05-22	PASS	Healthy (prior failure was workflow file syntax, not tests)
Linters	2026-05-22	PASS	Healthy
Upstream PyTorch Tests	2026-05-22	PASS	Healthy
Nightly Tests	2026-05-22	FAIL	Persistently failing since Apr 24 (28 days)

Nightly Failure Triage

Failure 1 & 2: aten.cat with zero-sized tensors — REGRESSION (NEW) Tests: test_cat_4d_dim2_zero, test_cat_4d_dim_m2_empty_first Error: InductorError: LoweringException: AssertionError — dim out of range for 1D empty tensor

Root cause: PR #2089 (faebd99) changed scalar/zero-sized tensor layout to 1D [0] instead of preserving original ndim. Cat lowering asserts $0 \leq \text{dim} < \text{ndim}$ which fails for the degenerate input.

Duration: 1 day (first appeared May 22)

Assessment: **Regression — needs fix in cat lowering or tensor layout**

Failure 3: layernorm_2d_lx_planning_reduction — FLAKY

Test:

LxPlanningTwoOpReductionTest::test_layernorm_2d_lx_planning_reduction

Error: AssertionError: compiled spyre <-> cpu mismatch — 1/128 elements, abs diff 0.156 vs 0.1 tolerance

Root cause: Borderline fp16 precision in LX planning reduction mode. Passed May 20–21, failed May 22.

Duration: Intermittent (1 day)

Assessment: **Flaky — tolerance too tight for fp16 + LX planning**

Overall nightly: 3 failed, 2240 passed, 62 skipped, 322 xfailed. Previous failures (mish tolerance, MRO test) are now fixed.

Version Alignment

Check	Expected	Actual	Status
PyTorch	torch~=2.11.0 (pyproject.toml)	2.11.0+cpu	MATCH

Repo Drift (last 7 days)

Repo	Commits	API-Touching	Risk
deeptools	63	10 (headers, CMake, dialects)	HIGH
flex	12	9 (FlexAllocator, MemoryType, C++20 upgrade)	HIGH
torch-spyre	71	—	—

Key upstream changes:

- flex#1070 — Remove default directive from FlexAllocator::allocate
- flex#1008 — Add MemoryType to FlexAllocator
- flex#1050 — Upgrade to C++20
- flex#1032 — Make AllocationDirective required (aligned with torch-spyre #1927)
- deeptools — DXP senprog.json, HCC lowering, Agen/dialects refactor

Blocking PRs

PR	Title	Why Blocking	Review Status
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#2218	Upgrade to PT 2.12	Major upstream version bump — blocks all downstream compat	REVIEW REQUIRED
#2190	Add memory type for allocate function	Allocator API change — requires deeptools/flex coordination	REVIEW REQUIRED
#2135	Add senprog handling	New program handling interface — cross-component	REVIEW REQUIRED

Project Velocity

Metric	Value	Notes
PRs merged (7 days)	50	Extremely high — testing infra, new ops, matmul/attention fixes
PRs in review	27 (non-draft)	14 awaiting first review, 3 approved, 4 changes requested
Approved (ready to merge)	3	#2219 flash attention test, #1984 chunk large ops, #2128 pytest-xdist
Stalled (>7 days)	17	Oldest: #1662 (36 days), #1853 (23 days), #1895 (22 days)

Open Bugs

Metric	Value
Total open	30
New (last 24h)	2 — #2222 logical_not on bool, #2207 Mistral-Small unsupported stick expr
Trend (7 days)	+6 filed, -3 closed = net +3
Critical / High Priority	None labeled

Smoke Test

Test	Status	Details
Import & device detection	PASS	Spyre available, 1 device, torch 2.11.0+cpu
Eager execution (4×4 add)	PASS	shape=[4, 4], fp16
Compiled execution (128×128 add)	PASS	torch.compile(), shape=[128, 128], fp16
Model E2E (Qwen3 0.6B)	PASS	5 tokens generated: " Paris. The capital of" — 3.7s load, 80.2s gen

Cross-Repo API Mismatches

NONE DETECTED — full stack built successfully despite high upstream drift. The `AllocationDirective` change (flex#1032) is aligned with torch-spyre#1927 (already merged).

Action Items

- 1 **[P1] Fix nightly regression — cat with zero-sized tensors.** PR #2089 introduced a layout change that breaks `aten.cat` lowering when inputs include empty tensors. Fix in `cat` lowering to handle `ndim` mismatch, or preserve original `ndim` in zero-sized tensor layout.
- 2 **[P1] Fix flaky layernorm tolerance.** Bump `atol` from 0.1 to 0.15 for `test_layernorm_2d_lx_planning_reduction` or use fixed seed. Borderline fp16 precision under LX planning.
- 3 **[P3] Review PT 2.12 upgrade PR #2218.** Blocking PR — version upgrades affect the entire team. Needs priority review.
- 4 **[P4] Merge 3 approved PRs.** #2219 (flash attention test), #1984 (chunk large ops), #2128 (pytest-xdist restart).
- 5 **[P5] Triage 17 stalled PRs.** Several are 20–36 days old. Close abandoned ones, ping authors on viable ones.